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EXP-3

TITLE: Finding ORF using a given DNA sequence

AIM: To find the ORF from the given DNA sequence and to find the gene name, introns and exons by using GENSCAN, DOTPLOT and other bioinformatic tools.

THEORY: In bioinformatics **GENSCAN** is a program to identify complete gene structures in genomic DNA. It is a GHMM-based program that can be used to predict the location of genes and their exon-intron boundaries in genomic sequences from a variety of organisms.

In bioinformatics, **BLAST (basic local alignment search tool)** is an algorithm and program for comparing primary biological sequence information, such as the amino-acid sequences of proteins or the nucleotides of DNA and/or RNA sequences. A BLAST search enables a researcher to compare a subject protein or nucleotide sequence (called a query) with a library or database of sequences, and identify database sequences that resemble the query sequence above a certain threshold.

In bioinformatics a **Dot Plot** is a graphical method for comparing two biological sequences and identifying regions of close similarity after sequence alignment. It is a type of recurrence plot.

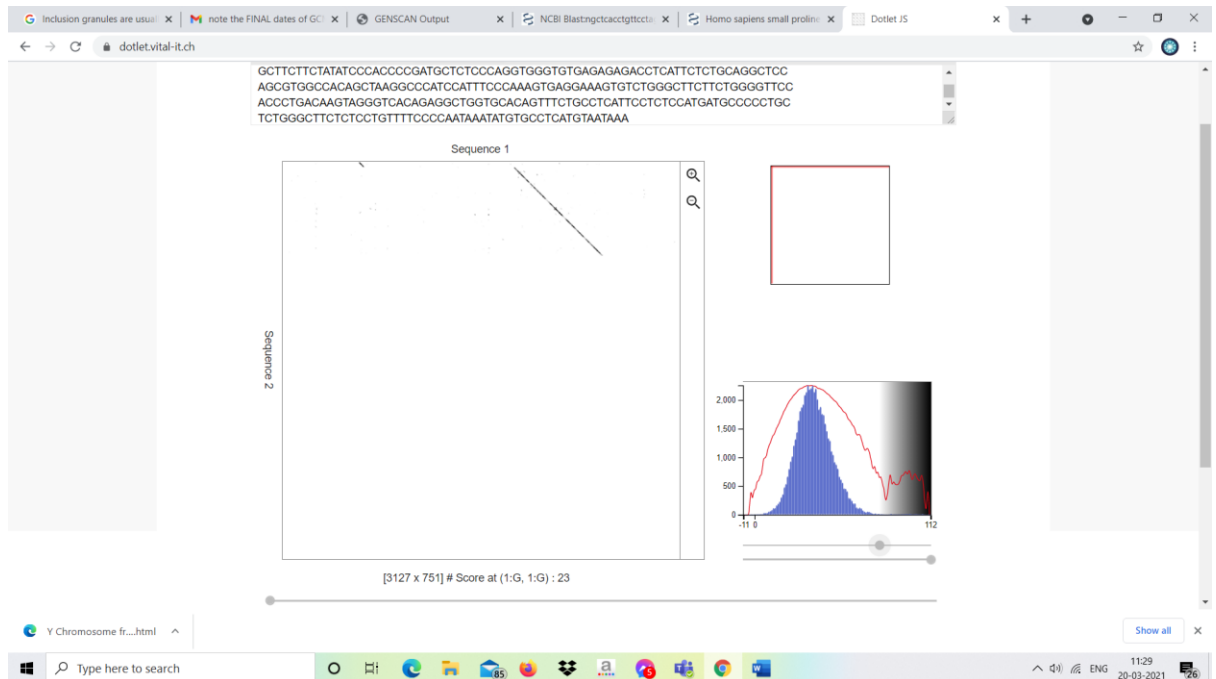
In eukaryotic gene, coding region or exon are present interrupted by non-coding region or intron which are many folds larger in sequence length. The entire gene is transcribed to make pre-mRNA which is then spliced to make mature mRNA.

PROCEDURE:

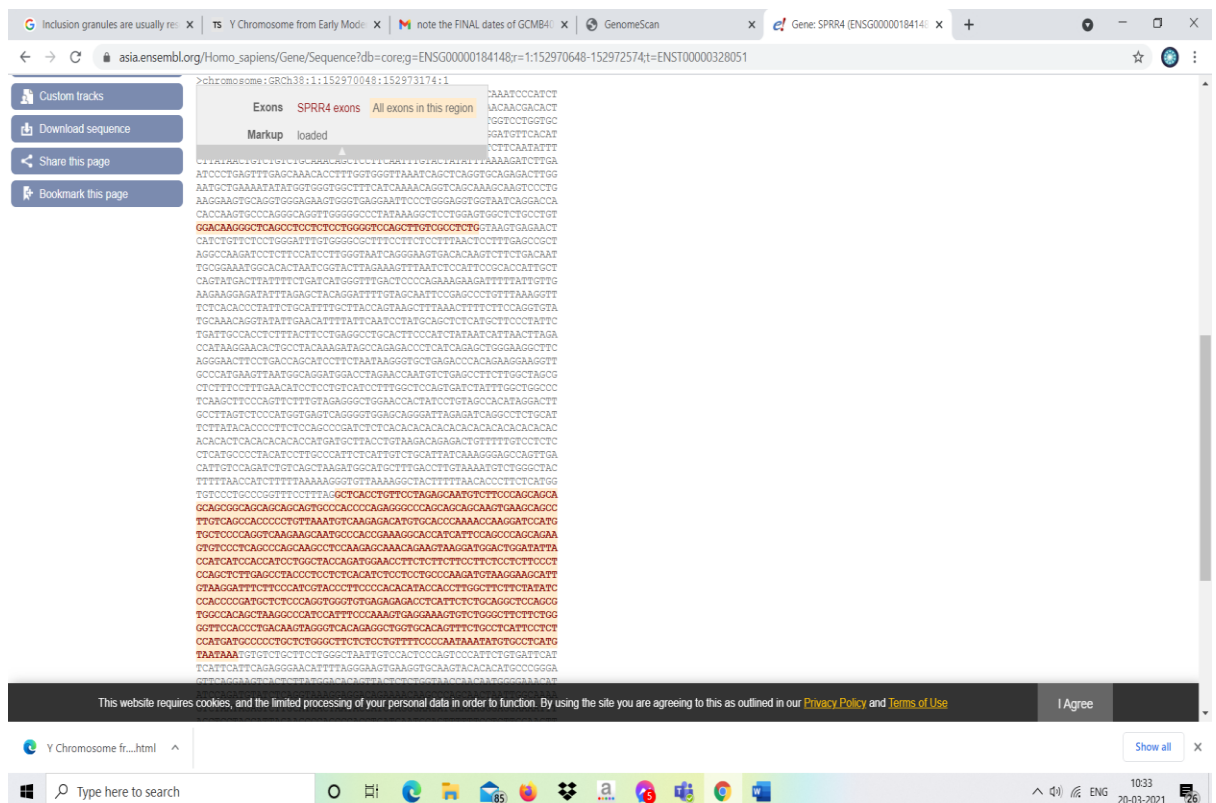
- ❖ First, we have to visit the Genscan website which is given by (www.hollywood.mit.edu).
- ❖ Then we have to paste the given DNA sequence in the dialog box given in the website, we also have to change “predicted peptides only” to “predicted CDS and peptide”.
- ❖ We have to click search Genscan then a DNA sequence coding for peptide will appear as an output.
- ❖ We have to copy the last sequence and paste it in the blast website.
- ❖ We have to open nucleotide blast website (blast.ncbi.nlm.nih.gov). Here in the query sequence box we have to give a greater than sign followed by the sequence copied from genscan and click on BLAST.
- ❖ After clicking we will find the description list from where we will select the SPRR4 proline rich protein. After that the pairwise sequence of premature DNA and mature mRNA sequence of SPRR4 will appear.
- ❖ After that we have to open the fasta format of the sequence and download it.
- ❖ Then we have to open the Dot Plot website (dotlet.vital-it.ch). In this website two sequence are given sequence 1 and sequence 2. In sequence 1 (x axis) we have to paste the first pre mRNA sequence and in sequence 2 (y axis) we have to paste the fasta format sequence of mature mRNA. As a result, a curve and graph was obtained.
- ❖ We then open the ensemble browser and opened the SPRR4 gene and then analysis of the gene sequence output was done to find the exon and intron position of SPRR4 and compared with the Dot Plot results.

OBSERVATIONS AND RESULTS: The results and observations are given below

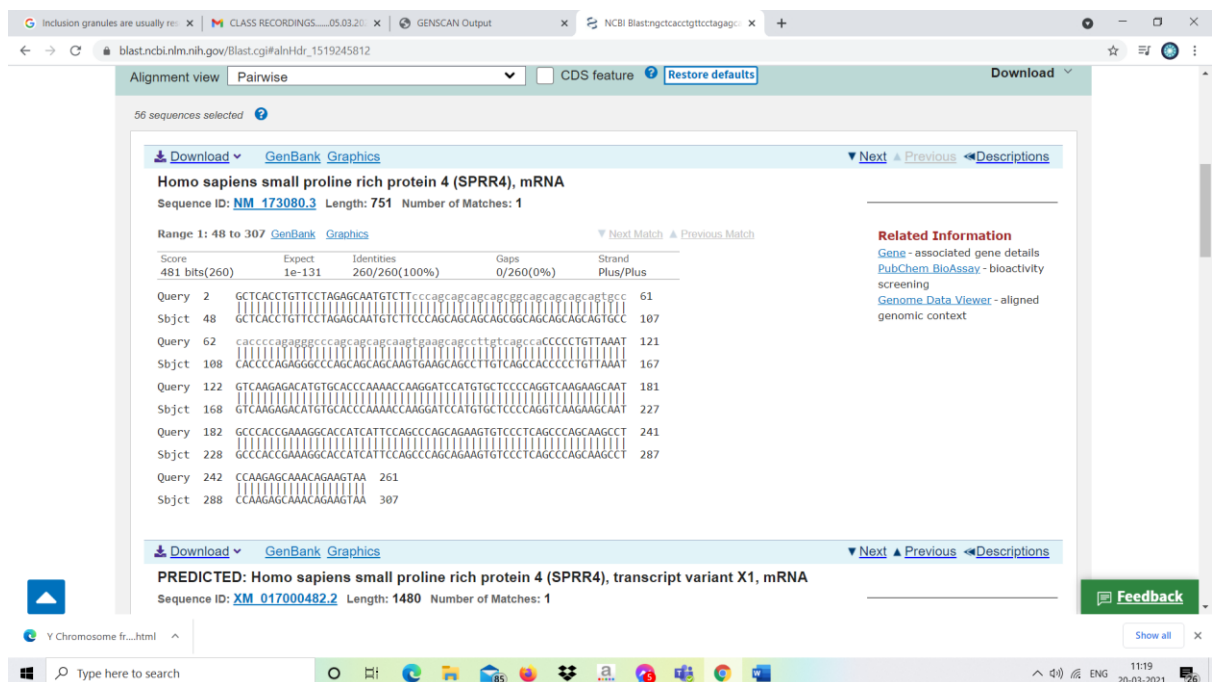
This figure shows the gene information of SPRR4 and it is obtained by BLAST search.



The figure shows the Dot Plot method which shows graphical representation of two sequences the pre mature mRNA and the mature mRNA sequence and it also shows the locations of exons.



This figure is obtained by Ensembl browser where exons are shown by red colour and introns by grey colour.



This figure shows the pairwise alignment of premature DNA and mature mRNA sequence of SPRR4.

CONCLUSION: We have found out the ORF of the given sequence and also the introns/exons by using Ensembl, Genscan, DotPlot and Blast.